MATH 101

Chapter 7 Homework

Systems of equations and inequalities

Sections 7.1 and 7.3 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Give exact answers. Use a basic four-function calculator only if needed; show algebraic reasoning.

Linear systems

1. Solve by substitution and check the ordered pair in both equations.

$y = 2x - 1, \qquad x + y = 8$

2. Solve by elimination and check both equations.

$2x + 3y = 7, \qquad 3x - 2y = 4$

3. Solve each system and describe its full solution set.

a) $x + 2y = 4, \quad 2x + 4y = 8$

b) $x + 2y = 4, \quad 2x + 4y = 10$

Nonlinear intersections

4. Solve algebraically, then sketch to confirm the number of intersections.

$y = x + 2, \qquad y = x^2$

5. Solve and check both ordered pairs.

$x^2 + y^2 = 25, \qquad y = x + 1$

6. Solve. Explain graphically why the solution count differs from Problem 4.

$y = x^2, \qquad y = -1$

7. Solve. Explain graphically why there is only one intersection.

$y = x^2, \qquad y = 2x - 1$

Regions and complete solutions

8. Sketch the solution region. Show solid or dashed boundaries and shade only the overlap.

$y \geq x^2, \qquad y < x + 2$

9. Sketch the solution region and decide whether (0, 0), (1, 1), and (2, 2) belong to it.

$x^2 + y^2 \leq 4, \qquad y \geq x$

10. For x squared + y squared = 25 and y = x + 1, a student reports x = 3 and x = -4 as the final answer. Explain what is missing; write and check the complete solutions.